Introduction to Database Systems (CSD202)

Lab Assignment#4

Date: 08/02/19

Field	Column Definition
Emp_ID	int-primary key
Name	Varchar(30)-not null
Department	Varchar(30)
Gender	Varchar(2)
Salary	Decimal(10,2)

Emp_ID	Name	Department	Gender	Salary
1	Reyan Fert	HR	M	80000
2	Luther Warn	Finance	M	70000.5
3	Audrey Joe	Purchase	F	46000.4
4	Kim John	Finance	F	30000
5	John Methew	Logistics	M	25000
6	Mery Lee		F	40000
7	Alex Bert	Finance	M	55000
8	Johnny W	Purchase	M	50000
9	Julia Ronald		F	99000.5
10	Sandra S	Logistics	F	20000.4

Creating Table Employee

```
mysql> create table employee
    -> (
    -> emp_id int,
    -> name varchar(30) NOT NULL,
    -> department varchar(30),
    -> gender varchar(2),
    -> salary decimal(10,2),
    -> PRIMARY KEY(emp_id)
    -> );
Query OK, 0 rows affected (0.05 sec)

mysql> describe employee;
+------------+---------------+------+-----+---------+-------+
| Field      | Type          | Null | Key | Default | Extra |
+------------+---------------+------+-----+---------+-------+
| emp_id     | int(11)       | NO   | PRI | NULL    |       |
| name       | varchar(30)   | NO   |     | NULL    |       |
| department | varchar(30)   | YES  |     | NULL    |       |
| gender     | varchar(2)    | YES  |     | NULL    |       |
| salary     | decimal(10,2) | YES  |     | NULL    |       |
+------------+---------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

1. Display the table employee.

```
mysql> select * from employee;
+--------+--------------+------------+--------+----------+
| emp_id | name         | department | gender | salary   |
+--------+--------------+------------+--------+----------+
|      1 | Reyan Fert   | HR         | M      | 80000.00 |
|      2 | Luther Warn  | Finance    | M      | 70000.50 |
|      3 | Audrey Joe   | Purchase   | F      | 46000.40 |
|      4 | Kim John     | Finance    | F      | 30000.00 |
|      5 | John Methew  | Logistics  | M      | 25000.00 |
|      6 | Mery lee     | NULL       | F      | 40000.00 |
|      7 | Alex Bert    | Finance    | M      | 55000.00 |
|      8 | Johnny W     | Purchase   | M      | 50000.00 |
|      9 | Julia Ronald | NULL       | F      | 99000.50 |
|     10 | Sandra S     | Logistics  | F      | 20000.40 |
+--------+--------------+------------+--------+----------+
10 rows in set (0.00 sec)
```

2. Number of employee from every department.

```
mysql> select department,count(emp_id) as No_Of_Employees from employee
    -> group by department;
+------------+-----------------+
| department | No_Of_Employees |
+------------+-----------------+
| HR         |               1 |
| Finance    |               3 |
| Purchase   |               2 |
| Logistics  |               2 |
| NULL       |               2 |
+------------+-----------------+
5 rows in set (0.00 sec)
```

3. Increase (in a single update query) the current salary of the employees as mentioned below:

   For HR dept- 20%
   For Finance- 15%
   For Purchase- 18%
   For logistics- 16%
   Display the table employee after increment.

```
mysql> update employee
    -> set salary =
    -> CASE
    -> WHEN department = 'HR' THEN salary*1.20
    -> WHEN department = 'Finance' THEN salary*1.15
    -> WHEN department = 'Purchase' THEN salary*1.18
    -> WHEN department = 'Logistics' THEN salary*1.16
    -> ELSE salary
    -> END;
Query OK, 8 rows affected, 3 warnings (0.01 sec)
Rows matched: 10  Changed: 8  Warnings: 3

mysql> select * from employee;
+--------+--------------+------------+--------+----------+
| emp_id | name         | department | gender | salary   |
+--------+--------------+------------+--------+----------+
|      1 | Reyan Fert   | HR         | M      | 96000.00 |
|      2 | Luther Warn  | Finance    | M      | 80500.58 |
|      3 | Audrey Joe   | Purchase   | F      | 54280.47 |
|      4 | Kim John     | Finance    | F      | 34500.00 |
|      5 | John Methew  | Logistics  | M      | 29000.00 |
|      6 | Mery lee     | NULL       | F      | 40000.00 |
|      7 | Alex Bert    | Finance    | M      | 63250.00 |
|      8 | Johnny W     | Purchase   | M      | 59000.00 |
|      9 | Julia Ronald | NULL       | F      | 99000.00 |
|     10 | Sandra S     | Logistics  | F      | 23200.46 |
+--------+--------------+------------+--------+----------+
10 rows in set (0.00 sec)
```

4. Display after update the gender of employee from F to M & M to F in a single update query.

```
mysql> update employee
    -> set gender =
    -> CASE
    -> WHEN gender = 'M' THEN 'F'
    -> WHEN gender = 'F' THEN 'M'
    -> END;
Query OK, 10 rows affected (0.01 sec)
Rows matched: 10  Changed: 10  Warnings: 0

mysql> select * from employee;
+--------+--------------+------------+--------+----------+
| emp_id | name         | department | gender | salary   |
+--------+--------------+------------+--------+----------+
|      1 | Reyan Fert   | HR         | F      | 96000.00 |
|      2 | Luther Warn  | Finance    | F      | 80500.58 |
|      3 | Audrey Joe   | Purchase   | M      | 54280.47 |
|      4 | Kim John     | Finance    | M      | 34500.00 |
|      5 | John Methew  | Logistics  | F      | 29000.00 |
|      6 | Mery lee     | NULL       | M      | 40000.00 |
|      7 | Alex Bert    | Finance    | F      | 63250.00 |
|      8 | Johnny W     | Purchase   | F      | 59000.00 |
|      9 | Julia Ronald | NULL       | M      | 99000.00 |
|     10 | Sandra S     | Logistics  | M      | 23200.46 |
+--------+--------------+------------+--------+----------+
10 rows in set (0.00 sec)
```

5. Display the list of those employees who belongs to any department.

```
mysql> select * from employee
    -> WHERE department != 'NULL';
+--------+-------------+------------+--------+----------+
| emp_id | name        | department | gender | salary   |
+--------+-------------+------------+--------+----------+
|      1 | Reyan Fert  | HR         | F      | 96000.00 |
|      2 | Luther Warn | Finance    | F      | 80500.58 |
|      3 | Audrey Joe  | Purchase   | M      | 54280.47 |
|      4 | Kim John    | Finance    | M      | 34500.00 |
|      5 | John Methew | Logistics  | F      | 29000.00 |
|      7 | Alex Bert   | Finance    | F      | 63250.00 |
|      8 | Johnny W    | Purchase   | F      | 59000.00 |
|     10 | Sandra S    | Logistics  | M      | 23200.46 |
+--------+-------------+------------+--------+----------+
8 rows in set (0.00 sec)
```

6. Display the list of those employees who currently do not have any department.

```
mysql> select * from employee
    -> WHERE department is NULL;
+--------+--------------+------------+--------+----------+
| emp_id | name         | department | gender | salary   |
+--------+--------------+------------+--------+----------+
|      6 | Mery lee     | NULL       | M      | 40000.00 |
|      9 | Julia Ronald | NULL       | M      | 99000.00 |
+--------+--------------+------------+--------+----------+
2 rows in set (0.00 sec)
```

7. Display the information of 3rd highest paid employee.

```
mysql> select * from employee ORDER BY salary DESC LIMIT 3,1;
+--------+-----------+------------+--------+----------+
| emp_id | name      | department | gender | salary   |
+--------+-----------+------------+--------+----------+
|      7 | Alex Bert | Finance    | F      | 63250.00 |
+--------+-----------+------------+--------+----------+
1 row in set (0.00 sec)
```

8. Drop the primary key from the table.

```
mysql> alter table employee
    -> drop PRIMARY KEY;
Query OK, 10 rows affected (0.09 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> describe employee;
+------------+---------------+------+-----+---------+-------+
| Field      | Type          | Null | Key | Default | Extra |
+------------+---------------+------+-----+---------+-------+
| emp_id     | int(11)       | NO   |     | NULL    |       |
| name       | varchar(30)   | NO   |     | NULL    |       |
| department | varchar(30)   | YES  |     | NULL    |       |
| gender     | varchar(2)    | YES  |     | NULL    |       |
| salary     | decimal(10,2) | YES  |     | NULL    |       |
+------------+---------------+------+-----+---------+-------+
5 rows in set (0.01 sec)
```

9. Change the size of gender column to Varchar(3)

```
mysql> alter table employee
    -> MODIFY gender varchar(3);
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> describe employee;
+------------+---------------+------+-----+---------+-------+
| Field      | Type          | Null | Key | Default | Extra |
+------------+---------------+------+-----+---------+-------+
| emp_id     | int(11)       | NO   |     | NULL    |       |
| name       | varchar(30)   | NO   |     | NULL    |       |
| department | varchar(30)   | YES  |     | NULL    |       |
| gender     | varchar(3)    | YES  |     | NULL    |       |
| salary     | decimal(10,2) | YES  |     | NULL    |       |
+------------+---------------+------+-----+---------+-------+
5 rows in set (0.00 sec)

mysql>
```

10. Drop the name filed from the table.

```
mysql> alter table employee
    -> drop name;
Query OK, 0 rows affected (0.21 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> describe employee;
+------------+---------------+------+-----+---------+-------+
| Field      | Type          | Null | Key | Default | Extra |
+------------+---------------+------+-----+---------+-------+
| emp_id     | int(11)       | NO   |     | NULL    |       |
| department | varchar(30)   | YES  |     | NULL    |       |
| gender     | varchar(3)    | YES  |     | NULL    |       |
| salary     | decimal(10,2) | YES  |     | NULL    |       |
+------------+---------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

11. Delete all the information from the table in a single query & display the schema.

```
mysql> delete from employee
    -> WHERE salary>0;
Query OK, 10 rows affected (0.01 sec)

mysql> describe employee;
+------------+---------------+------+-----+---------+-------+
| Field      | Type          | Null | Key | Default | Extra |
+------------+---------------+------+-----+---------+-------+
| emp_id     | int(11)       | NO   |     | NULL    |       |
| department | varchar(30)   | YES  |     | NULL    |       |
| gender     | varchar(3)    | YES  |     | NULL    |       |
| salary     | decimal(10,2) | YES  |     | NULL    |       |
+------------+---------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

12. Drop table employee.

```
mysql> drop table employee;
Query OK, 0 rows affected (0.03 sec)
```